Code No: 124CQ **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, February - 2024
DATABASE MANAGEMENT SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	What is a database?	[2]
b)	Quote example for aggregation in ER model.	[3]
c)	What is an unsafe query?	[2]
d)	What is a trigger?	[3]
e)	Define functional dependency.	[2]
f)	Give example for spurious tuples.	[3]
g)	Define schedule.	[2]
h)	What are the benefits of multiple granularity?	[3]
i)	Define index.	[2]
j)	What is static hashing?	[3]

PART - B

(50 Marks)

2. How does database management system provide data abstraction? Explain in detail. [10]

OR

3. What is an integrity constraint? Discuss various types of integrity constraints for DBMS. [10]

4. Illustrate all primitive operators available in Relational Algebra for data accessing. [10]

OR

5. Consider the following database schema to write queries in SQL:
Employee(Eid, Ename, Ecity)
Company(Cno, Cname, Ccity)
Works_for(Eid, Cno, Salary)
a) Find the employees hailing from Kolkata.
b) Find the employees working for Wipro.
c) Find the average salary of employees working for IBM. [3+3+4]

6. Explain the need of schema refinement in database design with suitable examples. [10]

OR

7. Make a comparison of Boyce Codd normal form and Fourth normal form. [10]

8. Discuss the concept of serializability with suitable schedules. [10]

OR

9. Demonstrate the working of validation based protocol. [10]

10. Compare tree based indexing with hash based indexing. [10]

OR

11.a) Illustrate dense indexing.

b) Explain insertion operation on hash file. [5+5]

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